Illustration of a proposed model.

Storyboard: OP2 star collapse (toy model)

Illustrative visualization of the

Quantized Space-Ledger toy model

Approved board on record · G = 0.01 · ~38 s · release: op2_collapse_v2.mp4

F1 0–5 s (5s)

Setup

Radial-shell Pauli packing · G = 0.01 (model units)

Motion and timing are illustrative, not measured collapse dynamics.

F2 5–12s (7s)

Supported lump

N = 141 · half-mass density ≈ 1.93

Finite lump held by Pauli packing. Blast-spill sets outer extent.

F3 12–19s (7s)

Add mass

Mass added → N = 3853 · half-mass density ≈ 8.1

Core denser (~4×). Different mass states, not a time series.

F4 19–25s (6s)

Support cap

Mass added → N ≈ 4738

Chosen support cap exceeded.

F5 25–33s (8s)

Inward collapse

Core falls to the grid floor. Halo stays extended.

Illustrative inward pile-up.

F6 33–38s (5s)

After collapse

Mass added → N ≈ 46,127 · half-mass density ≈ 5,500

Toy model—not a stellar simulation.

Layout: 1920×1080 · all on-screen text ≥38 px · header / art / caption bands separated

Illustration of a proposed model.

Notes page — numbers, sources, caveats

Kept off the video frames. Approved board; release file is op2_collapse_v2.mp4

- Dataset: Pauli v4 radial-shell mean-field toy · G = 0.01 (model units) · not a 3D stellar/hydro sim
- Choices: 2 spin slots/level/node; no self-load; blast-spill = max-radius load-packing rule (not ejecta)
- Support cap: LCAP = 7 is a disclosed interpretation of 8-neighbour spill capacity (model choice)
- Past the chosen cap, other-particle congestion pulls mass inward; R_half → 1 (lattice floor)
- Point A (supported): N=141, ρ_half≈1.93 (exact 1.925…), R_half≈2.06, R_out=3
- Point B (near collapse): N=3853, ρ_half≈8.1 (exact 8.066…), R_half≈3.85, R_out=8
- Collapse threshold: N=4738 (data/pauli_deg4_G0.01_excerpt.json collapse_N; that row has R_half=1)
- Point C (~10× collapse N): N=46127, ρ_half≈5,500 (exact 5506…), R_half=1, R_out=18
- ρ_half = (N/2)/(4/3 π R_{half}^3). Outer/halo density stays ~1–2 (thin ground-level envelope).
- B/A half-mass density ≈ 4.2× at G=0.01 — not real planet→WD density matching (~10^5).
- No sustained white-dwarf R ∝ $M^{-1/3}$ branch in this scan.
- A/B/threshold/C are separate mass states: bridge on screen by explicitly adding mass (A→B→threshold→C).
- Motion/timing are illustrative — not measured collapse dynamics.
- Do not depict: blast-spill ejecta, supernova, bounce, nuclear burning, neutron phase, BH/horizon.
- Sources: provenance/pauli_massdens_SUMMARY.md + data/pauli_massdens_G0.01_excerpt.json;
- provenance/pauli_deg4_SUMMARY.md + data/pauli_deg4_G0.01_excerpt.json

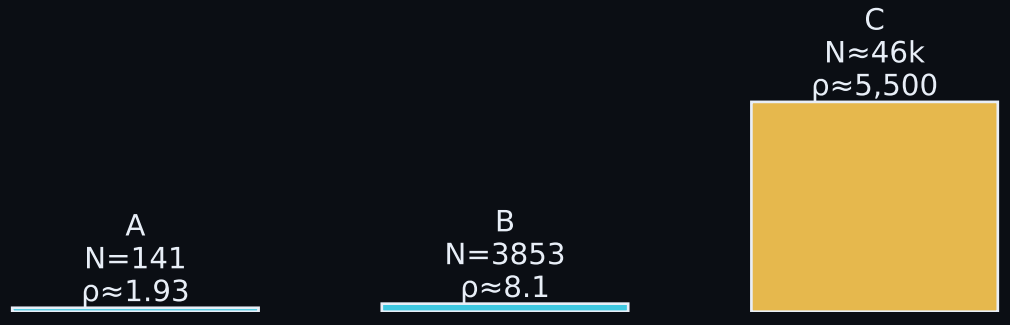
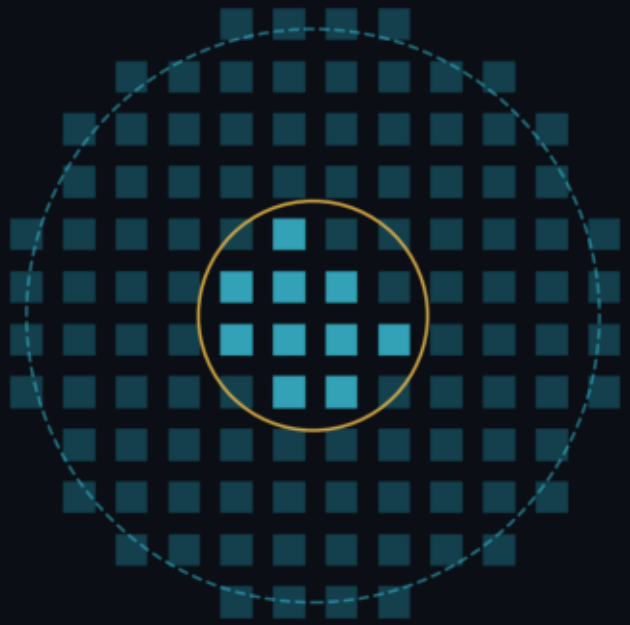


Illustration of a proposed model.

Illustrative visualization of the
Quantized Space-Ledger toy model

Setup



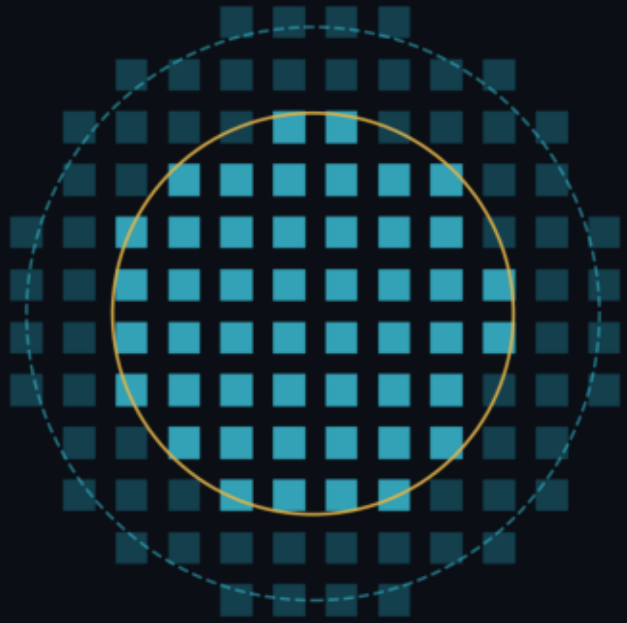
Radial-shell Pauli packing · G = 0.01 (model units)

Motion and timing are illustrative, not measured collapse dynamics.

Illustration of a proposed model.

Illustrative visualization of the
Quantized Space-Ledger toy model

Supported lump



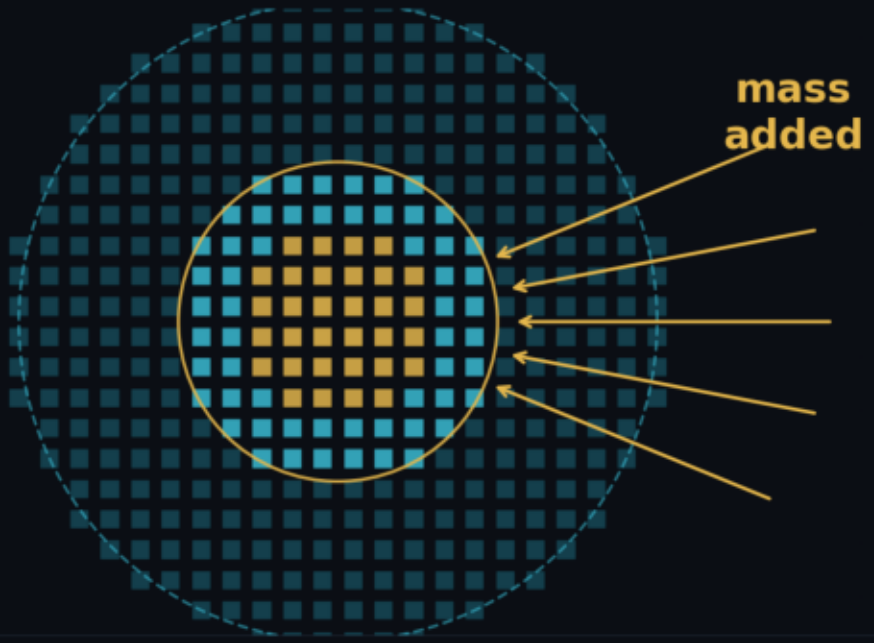
N = 141 · half-mass density ≈ 1.93

Finite lump held by Pauli packing. Blast-spill sets outer extent.

Illustration of a proposed model.

Illustrative visualization of the
Quantized Space-Ledger toy model

Add mass



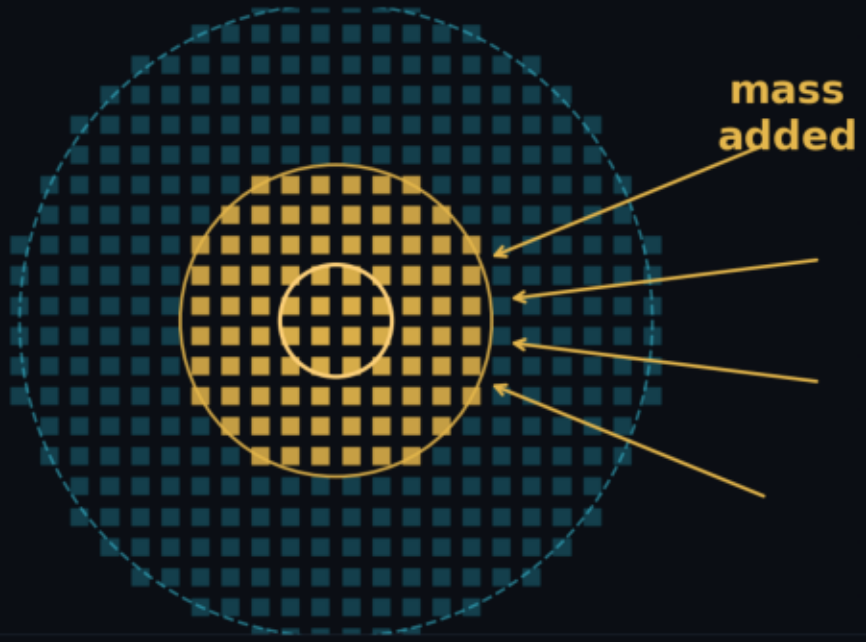
Mass added → N = 3853 · half-mass density ≈ 8.1

Core denser (~4×). Different mass states, not a time series.

Illustration of a proposed model.

Illustrative visualization of the
Quantized Space-Ledger toy model

Support cap



Mass added → N ≈ 4738

Chosen support cap exceeded.

Illustration of a proposed model.

Illustrative visualization of the
Quantized Space-Ledger toy model

Inward collapse



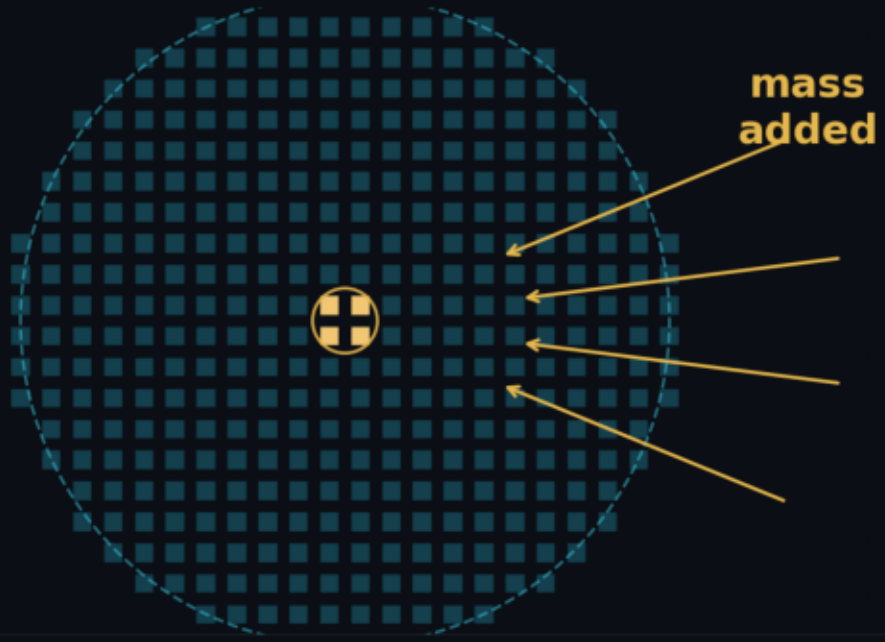
Core falls to the grid floor. Halo stays extended.

Illustrative inward pile-up.

Illustration of a proposed model.

Illustrative visualization of the
Quantized Space-Ledger toy model

After collapse



Mass added → N ≈ 46,127 · half-mass density ≈ 5,500

Toy model—not a stellar simulation.